

Vithana Accounting Automation

Integration Setup Guide

Gmail OAuth & Mail Ingestion · Zoho Books · QuickBooks Online

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1. Gmail OAuth Setup

Vithana uses Google OAuth 2.0 to securely authorise access to Gmail inboxes. Each company tenant has its own OAuth client credentials stored in the database — no shared credentials are used.

1.1 Create a Google Cloud Project

1. Go to **console.cloud.google.com** and sign in with a Google account that owns or manages the organisation.
2. Click the project selector at the top → **New Project**.
3. Name it (e.g., *vithana-prod*) and click **Create**.
4. Ensure the new project is selected in the top bar before continuing.

1.2 Enable the Gmail API

1. In the left menu navigate to **APIs & Services** → **Library**.
2. Search for **Gmail API** and click it.
3. Click **Enable**. Wait for the API to activate.

1.3 Configure the OAuth Consent Screen

1. Navigate to **APIs & Services** → **OAuth consent screen**.
2. Select **Internal** if users are within your Google Workspace org, or **External** for personal Gmail accounts.
3. Fill in **App name** (e.g., *Vithana Accounting*), **User support email**, and **Developer contact email**.
4. Under **Scopes** add: <https://www.googleapis.com/auth/gmail.readonly>
5. If **External**: under **Test users** add every Gmail address that will authorise during testing.
6. Save and continue until the consent screen is published.

■ Important

Until your app passes Google verification, only addresses listed as **Test users** can authorise. Add all developer and tester emails now.

1.4 Create OAuth 2.0 Credentials

1. Navigate to **APIs & Services** → **Credentials**.
2. Click **+ Create Credentials** → **OAuth client ID**.
3. Application type: **Web application**.
4. Add the authorised redirect URI: <http://<your-backend-host>/api/auth/google/callback>
5. Click **Create**. Download the JSON — it contains **client_id** and **client_secret**.

1.5 Register Credentials in Vithana Admin

1. Log in to the Vithana Admin Dashboard.
2. Navigate to **Companies** → <Company Name> → **OAuth Settings**.
3. Enter the **Client ID**, **Client Secret**, and **Redirect URI**.
4. Click **Save OAuth Config**.
5. The redirect URI must exactly match what you registered in Google Cloud.

Config Key	Description	Example / Notes
client_id	OAuth 2.0 Client ID from Google Cloud	1234567890-abc.apps.googleusercontent.com
client_secret	OAuth 2.0 Client Secret	GOCSPX-xxxxxxxxxxxx
redirect_uri	Must match Google Cloud exactly	https://api.vithana.com/api/auth/google/callback

2. Gmail Mail Ingestion (Pull)

Once OAuth credentials are configured, each user can authorise their Gmail mailbox. Vithana then pulls emails matching configured rules and extracts invoice attachments.

2.1 User Authorises Gmail Access

1. User logs in to the Vithana web app.
2. Navigate to **Settings** → **Email Sources** → **Connect Gmail**.
3. Click **Authorise Gmail**. A Google consent screen opens.
4. Grant **Read mail** permission.
5. On success the status shows **Connected** with the authorised email address.

Note

Vithana requests **read-only** Gmail access. It never sends, deletes, or modifies emails.

2.2 Configure Ingest Rules

Rules control which emails are picked up. Navigate to **Settings** → **Rules** and create rules:

- **Sender filter** — e.g., `invoices@vendor.com`
- **Subject keyword** — e.g., `invoice, bill`
- **Attachment type** — PDF (default), PNG, JPEG
- **Label/folder** — restrict to a specific Gmail label

2.3 Trigger a Manual Sync

1. Navigate to **Dashboard** → **Email Sources**.
2. Click **Sync Now** on the Gmail source.
3. Vithana fetches matching emails since the last sync timestamp.
4. Extracted attachments appear in **Invoices** → **Pending**.

2.4 Automated Polling

The backend scheduler polls Gmail automatically at the interval set in **Settings** → **System Config** → **EMAIL_POLL_INTERVAL_MINUTES** (default: 30 minutes). No manual sync is needed in production.

2.5 Environment Variables

Config Key	Description	Example / Notes
GOOGLE_REDIRECT_URI	Callback URL for OAuth flow	<code>https://api.vithana.com/api/auth/google/callback</code>

GMAIL_POLL_INTERVAL_MINUTES	Auto-sync frequency (minutes)	30
ATTACHMENT_DIR	Local path to store downloaded files	/app/attachments

3. Zoho Books Integration

Vithana pushes approved invoice drafts to Zoho Books as Bills. Authentication uses Zoho OAuth 2.0 with a long-lived **refresh token**.

3.1 Create a Zoho OAuth App

1. Go to **api-console.zoho.in** (India region) or **api-console.zoho.com** (Global).
2. Click **Add Client** → **Server-based Applications**.
3. Set **Redirect URI** to any valid HTTPS URL (e.g., `https://api.vithana.com/oauth/zoho/callback`). This URI is only used once during the grant-token exchange.
4. Note the **Client ID** and **Client Secret**.

3.2 Generate a Grant Token (One-Time)

Open this URL in your browser while logged into Zoho Books:

```
https://accounts.zoho.in/oauth/v2/auth?scope=ZohoBooks.fullaccess.all&client_id=YOUR_CLIENT_ID&response_type=code&redirect_uri=YOUR_REDIRECT_URI&access_type=offline
```

1. Approve the consent screen.
2. You are redirected to your redirect URI with `?code=1000.xxxxx` in the URL.
3. Copy that **code** value — it expires in 60 seconds.

3.3 Exchange Grant Token for Refresh Token

Run the following curl command immediately after copying the grant code:

```
curl -X POST https://accounts.zoho.in/oauth/v2/token \
-d "code=1000.YOUR_GRANT_CODE" \
-d "client_id=YOUR_CLIENT_ID" \
-d "client_secret=YOUR_CLIENT_SECRET" \
-d "redirect_uri=YOUR_REDIRECT_URI" \
-d "grant_type=authorization_code"
```

The response contains **refresh_token** — save it permanently. It does not expire unless revoked.

3.4 Get Your Organization ID

1. Log in to Zoho Books.
2. Navigate to **Settings** → **Organisation Profile**.
3. Copy the **Organisation ID** shown at the top.

3.5 Configure in Vithana

1. In the Vithana web app, navigate to **Settings** → **Integrations** → **Zoho Books** → **Configure**.
2. Fill in all required fields (see table below).
3. Click **Test Connection** to verify.

4. Click Save.

Config Key	Description	Example / Notes
client_id	Zoho OAuth Client ID	1000.ABCDEF...
client_secret	Zoho OAuth Client Secret	2842196c...
refresh_token	Long-lived refresh token from Step 3.3	1000.fd29a3...
organization_id	Zoho Books Organisation ID	12345678
base_url	API base (India: zohoapis.in, Global: .com)	https://www.zohoapis.in/books/v3
auth_url	Token endpoint	https://accounts.zoho.in/oauth/v2/token
default_account_id	Fallback Chart of Accounts ID (optional)	460000000000388
tax_id	Zoho tax rate ID for GST auto-split (optional)	460000000053001

3.6 GST Tax Handling

Vithana uses **Option A**: the `tax_id` is set on each bill line item and Zoho automatically computes CGST / SGST / IGST based on its configured tax rates. To find your `tax_id`:

- In Zoho Books go to **Settings** → **Taxes** → **Tax Rates**.
- Click the desired GST rate (e.g., GST 18%).
- The URL contains the tax ID: .../taxrates/**460000000053001**

Note

If `tax_id` is left blank, bills are pushed without GST. Zoho will use its own default tax rules if configured.

4. QuickBooks Online Integration

Vithana can push approved bills to QuickBooks Online (QBO) using the Intuit OAuth 2.0 flow. This section covers app creation, credential exchange, and Vithana configuration.

4.1 Create an Intuit Developer App

1. Go to **developer.intuit.com** and sign in with your Intuit account.
2. Click **Dashboard** → **Create an app**.
3. Select **QuickBooks Online and Payments**.
4. Choose a name (e.g., *Vithana Accounting*) and select the **com.intuit.quickbooks.accounting** scope.
5. Click **Create app**.

4.2 Configure Redirect URIs

1. In the app dashboard go to **Development** → **Keys & credentials** (or **Production** tab for live).
2. Under **Redirect URIs** add: `https://api.vithana.com/api/auth/quickbooks/callback`
3. Save changes.
4. Note the **Client ID** and **Client Secret** from this page.

■ Sandbox vs Production

Intuit provides a **Sandbox** environment with test companies. Use sandbox credentials during development. Switch to **Production** credentials before go-live.

4.3 Authorise and Obtain Tokens

1. In the Vithana Admin Dashboard navigate to **Settings** → **Integrations** → **QuickBooks** → **Connect**.
2. Click **Authorise QuickBooks**. You are redirected to Intuit's consent screen.
3. Select the QuickBooks company to connect and click **Connect**.
4. Vithana exchanges the authorisation code for **access_token** and **refresh_token** automatically and stores them.
5. The **Realm ID** (company ID) is captured from the callback and saved.

4.4 Manual Token Exchange (CLI)

If you need to exchange tokens manually outside the UI:

```
# Step 1 - Build the authorisation URL https://appcenter.intuit.com/connect/oauth2
?client_id=YOUR_CLIENT_ID &redirect_uri=YOUR_REDIRECT_URI &response_type=code
&scope=com.intuit.quickbooks.accounting &state=random_state_string # Step 2 - Exchange
the code curl -X POST https://oauth.platform.intuit.com/oauth2/v1/tokens/bearer \ -H
"Authorization: Basic base64(CLIENT_ID:CLIENT_SECRET)" \ -H "Content-Type:
application/x-www-form-urlencoded" \ -d
```

```
"grant_type=authorization_code&code;=YOUR_CODE&redirect;_uri=YOUR_REDIRECT_URI"
```

4.5 Configuration Fields

Config Key	Description	Example / Notes
client_id	Intuit app Client ID	ABcd1234EFgh...
client_secret	Intuit app Client Secret	XYZ9876...
refresh_token	Long-lived refresh token (180-day expiry)	AB11...
realm_id	QuickBooks company ID (Realm ID)	1234567890
base_url	QBO API base URL	https://quickbooks.api.intuit.com
auth_url	Intuit token endpoint	https://oauth.platform.intuit.com /oauth2/v1/tokens/bearer
environment	sandbox or production	production

4.6 Token Refresh

QBO access tokens expire after **1 hour**. Refresh tokens expire after **180 days** (or 100 days of inactivity). Vithana automatically refreshes the access token before each API call. Re-authorisation is required only if the refresh token expires.

■ Important

Set a calendar reminder to re-authorise QuickBooks every **150 days** to avoid refresh token expiry in production.

Appendix A: Troubleshooting

Platform	Error / Symptom	Resolution
Gmail	Access blocked: App has not completed Google verification	Your Google account is not listed as a Test User. Go to Google Auth Platform → Audience → Test Users and add the email address. Only needed during development; production requires app verification.
Gmail	Connection failed: HttpError / stale token	The stored OAuth token is invalid or expired. In Vithana go to Settings → Email Sources, disconnect the Gmail source, and re-authorise by clicking Connect Gmail again.
Gmail	Wrong redirect URI port	The redirect URI in Google Cloud Console must exactly match what Vithana sends. Check GOOGLE_REDIRECT_URI in .env and update the Google Cloud Console entry to match.
Zoho	Invalid grant / refresh token revoked	The refresh token has been revoked (e.g., password change, session purge). Repeat Section 3.2–3.3 to generate a new grant token and exchange it for a fresh refresh token.
Zoho	Vendor not found on Zoho	Vithana could not find the vendor in Zoho Books. Create a Vendor Mapping in Settings → Vendor Mappings, mapping the alias to an existing Zoho contact, or create the vendor in Zoho first.
Zoho	Bill created but no GST applied	The tax_id field is not set in the Zoho integration config. Find the tax rate ID in Zoho Books → Settings → Taxes → Tax Rates and add it to the integration settings.
QuickBooks	AuthenticationFailed / Invalid token	The access token has expired (1-hour TTL). Vithana auto-refreshes — if this persists, the refresh token may have expired (180 days). Re-authorise via Settings → Integrations → QuickBooks.
QuickBooks	Company not found / Invalid Realm ID	The realm_id does not match the connected QBO company. Check the Realm ID in your QBO URL (https://app.qbo.intuit.com/app/homepage?token=) and update the integration config.

Support

For platform-specific issues consult the official documentation:

- **Google OAuth:** developers.google.com/identity/protocols/oauth2
- **Gmail API:** developers.google.com/gmail/api
- **Zoho Books API:** www.zoho.com/books/api/v3

- **QuickBooks API:** developer.intuit.com/app/developer/qbo/docs